

String Art (sample solution)

COSC 1210 (CS I) Fall 26 | Augustana U.

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1. String Art

Problem: Build the poster below in Python. Respect the following rules:

1. Top and bottom rows are one character repeated with *
2. Each line with text has to be indented by repeating single empty spaces

```
=====
      BAKE SALE
    Friday at 3 pm
      Room 204
=====
    12 dozen for $5
    SEE YOU THERE
```

2. Solution

2.1. What length does the poster have?

```
print(len("====="))
```

```
30
```

2.2. The top and bottom line of the poster

```
print("=" * 30)
```

```
=====
```

2.3. Counting the leading spaces

Find out how many empty spaces there are before the first letter in each line of text:

```
print("        BAKE SALE".find('B'))
print("    Friday at 3 pm".find('F'))
print("        Room 204".find('R'))
print("12 dozen for $5".find('1'))
print("    SEE YOU THERE".find('S'))
```

```
10
7
10
6
8
```

But to do this, you have to know what the first letter is that you're looking for. Better to use a statement that tells you when the empty spaces end.

Define a list that holds the text lines:

```
text = [
    "        BAKE SALE",
    "    Friday at 3 pm",
    "        Room 204",
    "12 dozen for $5",
    "    SEE YOU THERE"
]
```

Go through every line, count characters and return the number of characters that are empty (before the first word):

```
for line in text:
    count = 0
    for character in line:
        count+=1
        if not character == ' ':
            print(count-1)
            break
```

```
10
7
10
6
8
```

2.4. Part 1 - Build the poster

The last two text lines are not needed yet. Put the printing into a function so that the code does not have to be repeated:

```
def poster():
    print("=" * 30)
```

```
print(" " * 10 + "Bake Sale")
print(" " * 7 + "Friday at 3 pm")
print(" " * 10 + "Room 204")
print("=" * 30)
```

```
poster()
```

```
=====
      Bake Sale
    Friday at 3 pm
      Room 204
=====
```

2.5. Part 2 - Add a count line

"12 dozen for \$5" must not be one string but must be composed by numbers and text using the `str` function:

```
poster()
print(" " * 6 + str(12) + " dozen for $" + str(5))
```

```
=====
      Bake Sale
    Friday at 3 pm
      Room 204
=====
12 dozen for $5
```

2.6. Part 3 - Shout it

Add `SEE YOU THERE` starting from the lowercase string and turning it uppercase in your code:

```
poster()
print(" " * 6 + str(12) + " dozen for $" + str(5))
print(" " * 8 + "see you there".upper())
```

```
=====
      Bake Sale
    Friday at 3 pm
      Room 204
=====
12 dozen for $5
SEE YOU THERE
```

2.7. An alternative method: `str.center`

```
help(str.center)
```

Help on method_descriptor:

```
center(self, width, fillchar=' ', /)
```

Return a centered string of length width.

Padding is done using the specified fill character (default is a space).

```
def poster2():
    print("=" * 30)
    print("Bake Sale".center(30, ' '))
    print("Friday at 3 pm".center(30, ' '))
    print("Room 204".center(30, ' '))
    print("=" * 30)
    print((str(12) + " dozen for $" + str(5)).center(30, ' '))
    print("see you there".upper().center(30, ' '))

poster2()
```

```
=====
      Bake Sale
    Friday at 3 pm
      Room 204
=====
    12 dozen for $5
    SEE YOU THERE
```

Author: Marcus Birkenkrahe

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